

8. Knn:

① Min Max norm: $\frac{2-0}{10-0} = 0.2$ $\frac{80-0}{100-0} = 0.8$

Customer B: $\frac{9}{10} = 0.9$ $\frac{15}{100} = 0.15$

Customer C: $\frac{6}{10} = 0.6$ $\frac{40}{100} = 0.4$

③ Distance of customer C to A and to B

Distance from A
Scaled $d = \sqrt{(0.2-0.6)^2 + (0.8-0.4)^2}$
 $= \sqrt{(0.4)^2 + 0.4^2}$
 $= \sqrt{0.16 + 0.16} = \sqrt{0.32} = 0.565$

Distance to B
Scaled $d = \sqrt{(0.9-0.6)^2 + (0.15-0.4)^2}$
 $= \sqrt{0.3^2 + (-0.25)^2}$
 $= \sqrt{0.09 + 0.0625} = \sqrt{0.1525} = 0.39$

③ the new customer is closest to customer 13
the predicted class is yes

question 12: Breaking a tie:

you should make the K an odd number ~~&~~ for ex. 3, so one point is taken out and the tie is broken.